

Transcripts: Second evaluation round

Expert 1 at Hawk

Date: October 25th, 2024

Time: 13:00

Place: Microsoft Teams

00:00:00 – 00:28:17

Interviewer Perfect. So here we are in the third round of our interviews, and this session will focus on some specific updates. I'll share my screen here. Last time, as you may recall, we discussed the design principles, and since then, I've gathered feedback from everyone, including you. Based on this input, I refined the design principles. Here's a quick recap of those principles: adaptive learning and continuous improvement, user-centric development and alignment, comprehensive data integration and quality assurance, AI-driven with targeted rules, scalability and flexibility, transparent and user-friendly explainability, a balanced approach to fraud prevention and detection, and finally, bias mitigation and ethical decision-making. So, as a next step in my thesis, I focused on how to implement these design principles practically to test their effectiveness and relevance. To do this, I created two things. First, an enterprise architecture model focused on the technical implementation of AI-driven rules and balanced fraud prevention and detection. We'll concentrate on this today. Secondly, a prototype designed for users like compliance officers. This prototype centers on transparency and user-friendly explainability within a dashboard layout, illustrating how users could navigate the system. We'll briefly look at this, but primarily for its key relevance to managers and developers. Starting with the enterprise architecture model, I'll walk you through it quickly and would love to get your input along the way. In a classic architecture setup, we have several layers: the business layer, the technology layer, the application layer, and finally, the technology layer. Zooming into the business layer, you'll see the business actors like the FDS, Fraud Detection System, users, which would include us as business managers and compliance officers. There's also the FDS provider, in this case, Hawk. You can also see how these actors collaborate continuously, which supports the principle of continuous improvement, even post-implementation. In the next layer, the support process layer, we see two primary support processes. One is managing fraud rules, where we are responsible for keeping rules updated to meet current needs. The other is model governance, which involves monitoring, improving, and deciding on model updates and deployments. Here, compliance plays a key role in ensuring that any deployments align with our risk protocols, a point we discussed previously. Moving to the main process layer, the fraud detection handling process, we've divided it into two primary workflows: card fraud detection and fraud alert handling. The process flow is straightforward. Starting with real-time analysis, we capture relevant transaction data and screen it using predefined card rules. If these rules are breached, we block the transaction; if not, it proceeds to algorithm-based screening. Depending on whether thresholds are exceeded, the transaction is either blocked or approved. Following approval, there's a post-transaction analysis to flag any additional suspicious activity. Transactions that pass this stage without issues are automatically closed to reduce manual workload, informed by user and customer profile feedback. If any suspicion arises, the system generates an alert for further investigation. Finally, we determine if it's a fraud case, and if so, we proceed with further investigation until resolved. Non-fraud cases are closed. Do you have any questions or feedback on the business layer before we dive into the more application-focused aspects? #00:00:03#

Interviewee My only question is, do we actually have this real-time and post-hoc flow at amnis? I wasn't aware, do we actually have real-time analysis and post-hoc analysis processes in place? #00:06:13#

Interviewer No, not yet. It's not directly implemented as an amnis use case, it's really more of a blueprint based on the design principles to show how it could look. Currently, I believe we only have the rules and the post-hoc analysis in place, but not real-time analysis #00:06:32#

Interviewee OK. #00:06:37#

Interviewer Or we still don't have enough data available. So, as I mentioned, this is just an example with the FDS user as a reference, but it demonstrates how you could implement it based on the design principles we discussed. So, it's not fully operational yet. #00:06:58#

Interviewee Makes sense. The only thing is, we actually have some customers who use auto-close models in the real-time flow. Some do, some don't, and I'm not entirely sure what the dividing criterion is, but just as an example, that's something we have in place as well. #00:07:11#

Interviewer Oh, OK. Really interesting, really. Most of the time when I talk with people, especially data scientists, it's usually about the auto-close model in the post-hoc analysis, not in real-time. But obviously, as you mentioned, it could also be incorporated in real-time if needed. Good to know, very interesting. #00:07:27#

Interviewee It's really a decision that the customers have to make, and I think it depends a bit on the rule type. For example, with more clear-cut, black-and-white rules, like transactions from high-risk countries, they typically wouldn't want auto-close enabled. But for other types of rules, they tend to be more open to it. So yes, we definitely have customers who permit auto-closing on certain blocking rules. #00:07:57#

Interviewer Right. Thank you for that, it's a helpful note. This really shows that while auto-close features can help reduce manual work, it ultimately depends on the customer's preferences. And that ties back into our goal of making the system as customer-centric as possible. Now, moving on to the more interesting parts, I'll skip the flow details for now but will focus on the main components here. First, we have the fraud detection rules engine, where transaction data is checked against these rules. Then there's the core of the system, the fraud detection algorithms, which handle pattern recognition and anomaly detection and also assist in reducing false positives. Another key component is the fraud detection platform, which acts as both an interaction and resolution tool. This is where users can engage with the system, and it also gathers user feedback on cases, whether they were correctly identified as fraud or not, which is fed back to the fraud detection algorithm for ongoing learning. Supporting this are the application layer components. These include the data integration pipeline, responsible for collecting all relevant data such as transaction data, historical client data, and contextual information like IP addresses or device IDs. Then we have feature extraction, which is critical as it processes and transforms raw data into relevant features for the fraud detection algorithm, both in real time and from historical data. The monitoring system is also key here. It tracks all metrics and performs data quality checks to spot any issues, such as missing data or shifts, and also monitors for potential biases. And an aspect we've discussed extensively in terms of compliance requirements, we have a staged model update system, where new models can be trained and validated in a staging environment before any updates are deployed into production. This ensures control over threshold adjustments and minimizes risk by allowing thorough testing beforehand. Lastly, though not as central, we're showcasing scalability and flexibility by implementing a dynamic compute cluster for the fraud detection algorithm. During high-traffic times like Christmas, for instance, this environment can scale by dynamically adding or removing nodes based on demand. That covers a lot, so I hope this overview was helpful. Now, I'd love to hear what you think, do you feel the design principles are well implemented here? Are there any features or aspects you feel are missing, or areas that could be improved? #00:08:06#

Interviewee I think it's very good. It's comprehensive and captures all the important elements. The only thing that comes to mind is with the feature extraction engine, it could also feed into the rules engine. In many cases, the features used in the rules and those used in the models overlap. So, it would make sense to compute these features only once and then send them to both the rules engine and the AI model. #00:12:25#

Interviewer That's a really good point. #00:13:11#

Interviewee But other than that, there was nothing else I noticed that I would change here. #00:13:14#

Interviewer The idea of having features flow to the rules engine makes sense. I hadn't considered that fraud detection rules would also need specific features to operate effectively. I always thought the algorithms needed them, but not the rules. It makes complete sense, though, that the data would also be necessary for the fraud detection rules to function properly. Thanks for pointing that out. #00:13:23#

Interviewee Exactly. #00:13:44#

Interviewer But is there anything else you'd like to add, or shall we move on to the other prototype? #00:13:49#

Interviewee I think we can move on. Everything looks good. #00:13:55#

Interviewer Thank you so much. As I mentioned, this is a prototype with simulated interactions, it's not fully functional yet. It's mainly to demonstrate user interaction. When you log in, you land on the dashboard, which we've discussed many times. I know the UX isn't polished since none of us are UX designers, but this gives a rough idea of how it might look. We've talked quite a bit about how users can monitor the fraud detection system. For instance, this dashboard includes customizable widgets that allow users to view relevant information based on their needs. Each user can configure it to include or exclude certain data, such as country information, to suit their requirements or organizational context. To keep track of system performance, we've included key metrics like accuracy, false positive rate, and false negative

rate, so users can get a snapshot of how the system is performing and see changes from the previous month. Additionally, system alerts notify users if something requires attention, for example, if the accuracy dips below a set threshold, like 96%, indicating they may need to contact developers or providers for adjustments. The dashboard layout can be tailored to different users' roles, a manager might prioritize metrics like overall accuracy, while a compliance officer might focus on transactional data, such as the number of transactions flagged, reviewed, blocked, or approved. There are also additional features, such as transaction export functions and the FDS performance overview, which we'll go over later. Managers or users can also manage the rules directly, adding or editing them as necessary. The model governance section includes documentation on ethical guidelines, details of model training, and data used, essentially a transparent log of all updates and changes. We also incorporated a change request and bug section. Here, users can create tickets, report issues, or request adjustments, further supporting collaboration between users and providers. That's a lot of information at once, I know, but what are your thoughts on the explainability and transparency aspects here? #00:13:58#

Interviewee One thing I think we should add is a section for something like an audit trail. #00:17:25#

Interviewer Audit trail, that's a good point. #00:17:32#

Interviewee Having an audit trail section would be essential, especially for compliance. It would allow us to track everything, any changes a user makes, such as adjustments to rule configurations or even actions as simple as viewing data. We'd log who accessed what data and include technical details like model outcomes. Apart from that, this looks great. I'd just suggest we add metrics specific to the false-positive reduction model or the auto-close model. These require close oversight, so having some dedicated widgets or metrics that show its performance would be useful to ensure it continues operating as intended. #00:17:35#

Interviewer That's a really good point. Not only did I overlook mentioning it, but I also forgot to incorporate something to monitor the auto-close alerts. Someone actually brought this up as well, a process where we periodically check random auto-close cases to ensure the system is still working correctly and that there's no unexpected behavior or false positives slipping through. This human oversight step is essential for validating the auto-close outcomes and ensuring quality control, which, as you said, is something we've discussed quite a bit but hasn't yet made it into this prototype. #00:18:49#

Interviewee Exactly. It would also be helpful to have a breakdown of this information by rule type. While the overall accuracy is important, as a user, I'd want to see which specific rules might not be performing well so I can target those for adjustments. #00:19:34#

Interviewer OK, that's interesting. So, are you suggesting showing a breakdown of which top rules are underperforming or creating the most alerts? Or did you mean highlighting the rules with the highest volume of alerts, regardless of performance? #00:20:02#

Interviewee Yes, in our system, for example, we have a dashboard that displays all configured rules along with how many alerts each one has generated. It's presented as a bar chart, with separate colors for false positives and true positives within each bar. This makes it easy to spot a rule that consistently produces a high number of incorrect alerts, if a bar is long but mostly the wrong color, it's a clear sign of an issue. We could even consider surfacing these problematic rules automatically as system alerts. That way, the system itself flags underperforming rules, reducing the need for users to manually interpret the bar chart. But yes, highlighting broken or problematic rules in some way would be important. #00:20:16#

Interviewer Exactly, that makes sense. When we talk about transparency and explainability, it's not just about the AI models themselves, it's about the entire system. This includes ensuring that all components, including the rules, are transparent and understandable. So, explainability should apply across the board, giving insight into not only how the AI makes decisions but also how each rule is performing and contributing to overall outcomes. #00:21:15#

Interviewee Exactly, and that's a great point. A seemingly simple but crucial metric is displaying the total number of transactions processed by the system. Many customers have regulatory requirements to prove that every transaction has been screened, so they need to verify our processed count against their own records. This helps ensure that every transaction has gone through screening without any losses or errors along the way. It's a straightforward addition but valuable for compliance and quality assurance. #00:21:42#

Interviewer Thanks. I'll quickly show an example of an alert here. For instance, this is a recent alert flagged by the AI. Without going too deep, you can see that the alert is clearly marked as blocked by the AI. We've incorporated transparency aspects like the AI confidence score, which indicates how confident the AI is that this particular card transaction is fraudulent. Additionally, we show feature impact, someone had mentioned SHAP values as a useful way to display which features contributed most to the decision. In this case, features like geolocation and timing are highlighted. For example, you can see on the map that most interactions typically occur in Europe, while this flagged transaction took place in Thailand. To keep it user-friendly, there's also a summary provided for managers who may not need the granular

details. It might say something like, blocked due to geolocation anomaly, this transaction in Thailand occurred 49 minutes after the last transaction in Switzerland, giving a quick overview of what triggered the alert. In previous discussions, we also talked about the integration of multiple data sources. This includes internal client data but also external data sources, so everything can be centralized. For instance, the system could pull relevant merchant data from the web, like flagging if the merchant has a high-risk rating or has been reported for cryptocurrency fraud. This way, the user doesn't need to switch between platforms to gather all relevant information, they won't have to access our internal database or Google additional details. It's all in one place, enhancing efficiency and decision-making. What do you think about that? #00:22:32#

Interviewee I think this is a fantastic screen, I really like it. I'd actually like to take a screenshot and bring some of these ideas back to our application. What stands out to me is that it feels like a power-user screen. There's a wealth of information here, yet it's very purposeful in surfacing exactly the right data for this alert. The geolocation graph, for instance, is a great touch. What I also appreciate is that you don't need many clicks to access all this relevant information. That's an issue we sometimes encounter in our Hawk system, where too many clicks are needed to reach key details. Here, though, it's direct and efficient. From my perspective, this is super strong, really well done. #00:24:45#

Interviewer As you can see, there's an option to click here for more details, it's not fully incorporated here, but the idea is to show a summary, with the option to dive deeper if needed. Moving to the next alert, you'll notice the view changes depending on the alert type. Here, it's a rule break, so it doesn't make sense to display an AI confidence level or a map. Instead, the system flags that it's related to a sanctioned country, displaying the list of sanctioned countries directly for context. External data could also be shown, like information from FATF indicating when Russia was added to the list, providing further context based on the specific case. For an auto-close case, the view shifts again. Here, we see that it's an auto-closed alert, with an explanation indicating that while the transaction took place in a high-risk country, it fit the customer's behavioral pattern based on their profile, so it was auto-closed. That's a quick walkthrough of the prototype. Do you feel this setup captures the transparency and case-specific details well? Or is there anything you feel is missing? What do you think?

Interviewee This looks great. The only other thought that might make it even more powerful is perhaps a widget for network analysis. It could help identify if involved parties are interconnected, or if a particular counterparty repeatedly links to different holding parties, for example. That's the only extra feature I can think of to take it up a notch, but overall, I think this is exceptionally well done, really impressive work. #00:27:21#

Interviewer Yes, I actually considered incorporating that as well. But if there's nothing else, I think we've covered the most important aspects of the user interaction prototype. If you don't have anything more to add, then we're all set. #00:27:55#

Interviewee Super, yes. #00:28:01#

Interviewer OK, thank you so much. #00:28:17#

Expert 2 at Hawk

Date: October 16th, 2024

Time: 14:30

Place: Microsoft Teams

00:00:00 – 00:25:54

Interviewer As previously mentioned, this is the third evaluation round, where we'll focus on evaluating specific concepts and prototypes. Our goal today is to test if the design principles we discussed in our last meeting, now refined based on your feedback, are practical, usable, and applicable in real-life scenarios. I would appreciate your expert opinion on this. Before diving into the enterprise architecture model to assess if these design principles can be implemented effectively, I'll quickly recap the key design principles for your reference, as it has been a while since our last discussion. We emphasized continuous learning and improvement, seamless development and alignment, comprehensive data integration and quality assurance, AI-driven and targeted rule design, as well as scalability and flexibility. Transparency and user-friendly explainability were also highlighted, along with a balanced approach to fraud prevention and protection, and finally, bias mitigation and ethical decision-making. While I've revised some of the titles based on your input, the core ideas remain the same. To demonstrate how these principles could be implemented in a real-world scenario, I created an enterprise architecture model. Now, let's go through this model. It consists of multiple layers. At the top, we have the business actors. For example, there's the FDS user, who interacts with the system, and on the right side, we have the fraud detection service provider, like Hawk, which supplies the fraud detection system. Next is the support process layer, which includes all the processes that assist in improving or managing the fraud detection system. Below that, we have the main business process layer, which is divided into two main areas, card fraud detection and fraud alert handling, both critical components of this system. Then, there is the application layer, which is more relevant here, as it contains the main applications for fraud detection as well as supporting applications that ensure smooth operations. Finally, the technical infrastructure layer shows the underlying technology that supports all these components. Are there any questions about these layers before we dive deeper into the specifics? #00:00:04#

Interviewee The production environment, could you clarify what exactly you mean? #00:03:50#

Interviewer We'll move on for now, as we will discuss it later. #00:03:55#

Interviewee Alright, no worries. #00:03:58#

Interviewer Let's go through each design principle to check if it's been implemented correctly. The first one is adaptive learning and continuous improvement. What I mean by that is the focus on ongoing improvement and learning. In this context, the FDS user and the provider collaborate continuously, not only during development but also after the system is live. They should still work together to enhance the system further. To support this, we have two key tasks. First, we have fraud rule governance, which involves managing the fraud rules in use. This is mostly done by the FDS user, which would be us, as we want to configure these rules in real-time, adjusting them according to sanctions or other regulations. The second, and larger, task is model governance, which involves managing the algorithms. This includes key processes such as improving and updating models, making decisions about deployments, and monitoring the models. This responsibility is shared between the provider and the user. We are both responsible for monitoring the performance of these models, but when it comes to improving and updating them, that mainly falls on your side as the provider. However, deployment decisions are made collaboratively. Any questions on this part? #00:04:04#

Interviewee That sounds good so far. #05:54#

Interviewer If it gets too complicated, please feel free to let me know. Describing such a model can be challenging, so if there's anything unclear or difficult to grasp, just let me know. #00:05:57#

Interviewee I'll let you know, yes. #00:06:08#

Interviewer If we move on to the business process layer, it's quite straightforward. The process is simple. First, we receive the card transactions and capture all the relevant data. This data is then screened against the fraud rules. For instance, if it's a card transaction from a sanctioned country, we don't need to analyze it further, it's automatically blocked. There's no need for further evaluation if the rule is breached. However, if there is no rule breach, we perform a preventive screening based on our fraud detection algorithms. If a certain threshold is exceeded and the AI identifies unusual patterns in the transaction, the card transaction is

blocked. If no suspicion is raised, the transaction is approved. Once approved, it moves on to the post hoc analysis, where all approved transactions are again screened by our fraud detection algorithms. If no threshold is exceeded, no further action is needed. If a threshold is exceeded, we check whether the case can be automatically closed based on feedback from the compliance officer or if it fits a pattern for similar transactions. If it can be automatically closed, the case is resolved. However, if it cannot be closed automatically, a fraud alert is created. In that case, the compliance officer investigates, provides feedback on whether it was indeed fraud, and specifies what type of fraud it was. This feedback helps improve the system. If the case is confirmed as fraud, further investigation takes place, or the fraud case is closed if it turns out to be a false positive. Any questions about this process? #00:06:10#

Interviewee Really, no questions. I have to say this looks very familiar to the implementation we have, so it looks good. #00:08:28#

Interviewer Perfect. Now, the more interesting part is how the application functions behind the scenes. Starting from the left side, we have the fraud detection rules, which are incorporated into a component that acts like a fraud rule machine. This machine checks each transaction against the established rules. The core part is the fraud detection algorithms, which are responsible for pattern recognition and anomaly detection. They analyze the transactions and help in reducing false positives. Another key component is the fraud detection interface itself. This is the platform, whether a web app or something else, that visualizes everything and allows users to work with the data. It also plays a crucial role in collecting user feedback, which is then fed back into the fraud detection algorithms. This feedback helps the system learn from past cases and improve its performance for future transactions, adapting and updating as necessary based on what it learns. Any questions or feedback on these main components? #00:08:35#

Interviewee That's exactly what I was looking for as well. In terms of validation, you mentioned fraud detection, and it might also be beneficial for the fraud detection mechanism to include a review process for some transactions that didn't trigger alerts. For example, if the system processes 100 transactions and only flags two as suspicious, it could still be valuable to randomly sample, say, two additional transactions from that batch, especially those above a certain threshold, like over 1000, just to verify that these unflagged transactions don't contain hidden issues that should have been reviewed. This concept ties into the auto-closed fraud cases you mentioned at the business level. When an AI model auto-closes a case, there should always be an opportunity for users to review these cases. It would be useful to occasionally sample some of those auto-closed cases, allowing users to verify that they were correctly handled, and that the AI made the right decision in closing them without further investigation. #00:09:56#

Interviewer Exactly. That is a really interesting point. What you are suggesting is that while the goal is to reduce manual work by minimizing false positives, there should still be a level of oversight and control over the system's decisions. So, even though the system is designed to handle much of the process autonomously, it would be important for compliance officers to periodically review a set of auto-closed cases. For example, the system could flag a few recent auto-closed cases for manual verification, allowing the compliance officer to check if they were indeed handled correctly. This ensures that the system remains accountable and that any potential issues are caught early. #00:11:08#

Interviewee Yes, exactly. Instead of purely random sampling, you could base the sampling on specific risk criteria. If a case was auto-closed, it's likely because it was initially flagged by a rule, meaning there were some underlying risk factors. For example, transactions with cross-border elements, high amounts, or other potential red flags could be prioritized for review. By sorting these cases based on their risk level, the system could provide a more targeted review process for compliance officers. This approach ensures that higher-risk transactions still receive some oversight while allowing the operators and compliance teams to maintain a clear understanding of what the AI is doing, helping them validate the system's effectiveness. #00:11:50#

Interviewer Great. So moving further down, we have the support application layer. This layer includes the data integration pipeline, which is responsible for collecting all the relevant data from various sources, like transaction data, client or historical data from our own database, as well as contextual data like IP addresses or device IDs. Essentially, it brings together data from different sources to ensure everything necessary is collected. Once the raw data is gathered, it goes through feature extraction, which pulls out all the relevant features needed for the fraud detection algorithm. This process ensures that the algorithm has all the important data points required for accurate fraud detection. In addition to this, there are two other crucial support applications. First, there is the monitoring system, which continuously checks key metrics related to performance, data quality, such as identifying missing data or shifts in data, and potential bias. It also tracks the input from compliance officers and ensures all significant elements are functioning correctly. Then, there's the staged model update system, which relates to your earlier question. This component allows you to carry out model testing, training, validation, feature engineering, and threshold optimization in a separate environment, essentially a staging environment. This ensures that everything is tested thoroughly before being deployed into the live system, where it interacts with real-time transactions. On the technical side, we also have a dynamic compute cluster that supports scalability. During high-transaction periods, the system can dynamically add nodes to handle the increased load, and when transaction volume is lower, it can scale down, freeing up resources. So, my

question to you is, given all this information, how do you feel about the design principles and their implementation? Do you think they are applicable and capable of functioning in real time, based on what you've seen so far? #00:12:26#

Interviewee Yes, I think those are definitely reflected well. For example, the staging environment is a great setup, and you could even have a separate environment dedicated solely to analysis, where all the models are developed. As long as the data moves through to production after the staging phase, that acts as the final confirmation. I see all the principles well represented here, and I think they're presented in a way that makes their implementation clear and effective. #00:15:07#

Interviewer Exactly, that's a crucial point. Any updates, particularly significant ones like threshold optimizations or major adjustments, should not go directly into the live environment without proper testing. While user feedback and transactional behavior can feed directly into the fraud detection algorithm to keep it adaptable, larger updates need to be tested in the staging environment first. Once they are validated and get approval from the users or from us, they can then be deployed to the live system. If there are no further remarks, I'll move forward. #00:15:40#

Interviewee Great, that's cool. #00:16:23#

Interviewer Perfect, thank you so much. Now, here's the prototype I developed for the user interaction. Keep in mind, we're not UX designers, so it's a bit basic, but it captures what we've discussed. A big focus has been on how to keep track of metrics, so I created a dashboard for the fraud detection system. This dashboard allows us to monitor all the important metrics in real-time, providing transparency and making it easier to track system performance. For example, the dashboard can show the accuracy, which might currently be at 95%, and whether it's improving. It can also display the false positive rate and other key statistics. You can customize the metrics, adding or removing them as needed. From your expert perspective, do you think this kind of dashboard is something you could work with? Is it essential or do you see anything here that might be redundant? #00:16:24#

Interviewee I agree, it's super important to have this kind of dashboard. If you could choose another variation, it might be useful to have one that displays the same information but tracks it over time. This would give you a clearer view of trends and help spot any long-term patterns or changes. That's often crucial for understanding the system's performance at a deeper level. #00:17:32#

Interviewer The next point is crucial, especially when it comes to tracking improvements over time. As you mentioned, it's not just about seeing metrics like accuracy at a single moment but also understanding how they evolve after updates are deployed. For example, being able to track when an update was implemented and seeing how performance metrics like accuracy were affected can be valuable. You could identify if an update caused a drop in accuracy, and then later, observe another update that brought the accuracy back up. Implementing this kind of functionality would be highly beneficial, giving users a clear view of the system's progression and the direct impact of updates. It adds transparency and allows for more informed decision-making. I think it's a great idea. What are your thoughts on it? #00:17:43#

Interviewee That's definitely a must on all those systems. #00:18:19#

Interviewer Another very valuable addition, I think, is having the ability to directly manage the fraud detection rules within the same system, would definitely enhance its functionality. Integrating a model governance part where users can see things like ethical guidelines, details about the training process, and a full documentation trail, such as a changelog for updates and changes, would provide a comprehensive view of how the system operates. Also, incorporating features for change requests or issue reporting is critical. For instance, if a user flags an alert that seems out of place, they can log it directly into the system, ensuring that there's always a clear line of communication and collaboration. This not only improves transparency but also helps continuously refine the system based on user feedback. Incorporating all of these features, rule management, model governance, and issue tracking, into one integrated system makes a lot of sense. It keeps everything centralized, improves cooperation between users and developers, and ensures that any anomalies or necessary adjustments are well-documented and acted upon. I think it would definitely add value to the system. What are your thoughts on that? #00:18:28#

Interviewee What exactly do you mean? #00:19:28#

Interviewer Right, I mean in the model governance section, you would also have areas to track open tickets, such as bugs or change requests sent to the provider. This would allow you to monitor issues or requests in real-time and ensure they are addressed properly. #00:19:32#

Interviewee The governance aspect is definitely a must. For instance, think about auditing transactions, we need to be able to track which model was used for a transaction that happened two years ago, especially if it later turned out to be part of a larger fraud. The ability to trace back and say, this transaction was evaluated using this specific model, and then have governance to show which risk factors were considered and what

data the model was analyzing is extremely important. I would say this is crucial and absolutely necessary here. #00:19:43#

Interviewer Thanks, moving forward, in the last part, when we look at an alert, we can see the details available for the user, but the transaction is blocked by the AI, so we know it was not approved. The reason might be something like a location anomaly. As we go further, we can also see that the AI's confidence level is displayed, let's say it shows that the AI is 91% confident this is a fraudulent card transaction. Additionally, we can see the feature impacts using SHAP values, which indicate which features had the most influence on the decision. For example, in this case, the key factors might be location and timing. It's just a showcase, but it demonstrates how these features impacted the decision. We can also see a map, where the cardholder usually performs transactions in Europe, but in this instance, it was in Thailand, which raises suspicion. Furthermore, the system integrates different data sources, such as the client profile, internal data, and external data, which shows how the fraud detection system works by pulling from multiple sources. What are your thoughts on these AI confidence and feature impact functionalities? #00:20:24#

Interviewee Yes, exactly. These are some of the features we have, and I think they are really beneficial for the industry. It's not just about flagging a transaction as anomalous but also explaining why the transaction was detected as fraudulent, showing which factors contributed to that decision, and the confidence level behind it. This kind of transparency is definitely a must. While feature impact might not always be mandatory, it's certainly very important. Having this level of explainability really sets one solution apart from another, and it plays a critical role when companies are deciding which solution to adopt. It's a key differentiator. #00:22:05#

Interviewer I mean, not all of the information needs to be discussed in detail, or perhaps it could be summarized. I think I spoke with someone else about this, showing that the system could also provide a short summary of an alert or a risk score analysis. These are all different components that contribute to creating more transparency. I've talked quite a bit, so I'd like to turn it over to you now. What do you think about everything we've discussed so far? Are these features presented in a way that makes sense, and do they align with the design principles we've laid out? What's your overall feeling? Do you think something is missing, or is there something that stood out to you as really good or not so good in your opinion? #22:55#

Interviewee For me, the design principles are really well illustrated. I think I see all of them represented, but there's one area that might need more attention, and that's data validation. Often, the model is functioning properly, the metrics look good, and everything seems to be in place. However, what can sometimes be missing is proper data validation. The entire transaction process relies on multiple data sources, and inconsistencies can arise. For example, someone might be sending country information in the A2 standard, but it was gathered from an A3 source. This kind of discrepancy could cause transactions to not be flagged correctly. It might be useful to introduce a general data validation step, where we can monitor the quality of incoming data values. For instance, we could track metrics like country count or field cardinality to ensure consistency. This way, if a particular field's values are constantly changing when they shouldn't be, or if there's an unexpected shift, it would be caught early. #23:39#

Interviewer Maybe I missed this earlier, but if we go back, we can see that the raw data goes to the monitoring system. There is the functionality to monitor data quality as well. Are you thinking more along the lines of a dashboard where you can continuously see the data that's coming in and track its quality in real-time, or were you referring to something else? #24:50#

Interviewee Oh, sorry about that. In the end, monitoring data quality might not be the responsibility of the fraud experts, but rather the people maintaining the system, right? I didn't notice it earlier, but if it's already implemented, then that should cover it. I think that's everything I saw. #00:25:04#

Interviewer One thing that I think is still missing, in my expert opinion, is the validation of random samples, particularly from the auto-closed cases. It's important to have control over these, ensuring that the system is functioning correctly. Regularly reviewing a sample of auto-closed cases would add an extra layer of oversight. If there's nothing else to add, then I think we're done. #00:25:34#

Interviewee I think it looks great overall, really a nice job. Honestly, well done. #00:26:00#

Interviewer Thank you so much. #00:26:03#

Expert 3 at Hawk

Date: October 15th, 2024

Time: 12:00

Place: Microsoft Teams

00:00:00 – 00: 27:20

- Interviewer** As I mentioned in the last round, this second evaluation round focuses on gathering feedback on the concepts and prototypes that were developed based on the design principles and the features derived from them. With you, I would like to start by discussing the first concept, the enterprise architecture model. This model illustrates how a fraud detection system for card transactions can or should be implemented within a company. As usual with our enterprise architecture models, we have multiple layers, and I will guide you through them. I will collect feedback on each layer. Starting with the business layer, we have the business actors. On one side, we have the FDS user, such as us, amnis, and on the other side, the FDS provider, which in this case is Hawk. These two actors work closely together, as shown by the collaboration relationship. As we've discussed many times, this is an ongoing partnership, not only during development but also throughout future iterations. Moving to the next layer, the support process layer, this contains processes that support the main fraud detection system. For example, we have the fraud rule governance function. This isn't directly related to the provider but is more aligned with the user's responsibilities. In this case, it's our job to keep track of the rules, adapting them to our needs. For instance, if there's a sanctioned country, we have a rule in place for that, and we need to maintain it. Meanwhile, the FDS provider is responsible for improving the model, applying updates, and ensuring it remains up to date and functioning well. This is the provider's role. I'm not sure if I've explained this clearly, as it's my first time describing such a model. If anything seems unclear or complicated, please feel free to ask questions. #00:00:04#
- Interviewee** What I understand is that the FDS provider is Hawk, and the FDS user is amnis, correct? #00:02:52#
- Interviewer** The FDS, or fraud detection system, provider is Hawk, correct. You are providing the platform, and the FDS user, in this case, is amnis. So from that perspective, we are the user, and you are the provider. As you can see, we also have the process for monitoring models. Do you have any questions about that? #00:02:57#
- Interviewee** Just a quick question. Why isn't the user involved in improving and updating models? I understand they wouldn't be involved in the actual coding, but the suggestions should definitely come from both sides. #00:03:41#
- Interviewer** That's a really good point. What I meant to show is that the actual improvement and updating of models needs to be done by the provider because, as you mentioned, they have the data scientists for that. #00:03:54#
- Interviewee** I think maybe you could add another node after the monitor models step. There could be a line connecting it to improve and update models. That way, after monitoring, the user can provide input based on their findings, like suggesting that certain patterns, such as round amounts or repeated amounts, should be checked. #00:04:07#
- Interviewer** That's perfect, thank you. It makes sense. In the end, both the user and the provider need to be involved in improving and updating the models. You're right, it's not just your task but a shared responsibility. As I mentioned, this is part of the support process, which includes governance and management of the fraud rules. Now, moving on to the main process, which is the fraud detection handling. This is split into two main parts, card fraud detection and fraud alert handling. So, let's discuss card fraud detection first. If we zoom in a bit, you can see that the process is divided into real-time analysis, where all incoming transactions are checked in real-time, and post-hoc analysis, which reviews transactions that have already been approved to identify any unusual behaviors or patterns. The process is quite straightforward. First, we receive the card transactions and capture all relevant data from various sources, historical client data, client specific data, contextual data, and transaction data. We then screen the transactions against the predefined rules. Since this is a hybrid approach, if there's a rule breach, for example, a transaction involving a sanctioned country, the transaction is blocked immediately. If there's no rule breach, the algorithm takes over and performs a preventive screening. If the AI determines that a threshold is exceeded and has high confidence that the transaction is fraudulent, the transaction is blocked. If not, it's approved and moves to the post-hoc analysis. In this step, even after approval, transactions are screened again. If nothing suspicious is found, the process ends there. However, if there is something suspicious, the system will try to auto-close the case based on previous user feedback or patterns. For example, if a

client frequently conducts business in high-risk countries, the system might automatically close the case to reduce manual work. If the system is highly confident that it's fraud and cannot close it automatically, further action is taken. #00:04:33#

Interviewee My concern is, why are you not using the fraud detection AI model? Why is it only rule-based as it currently stands? #00:07:43#

Interviewer I understand your concern. We'll get into the technical details in the application layer later. While we do screen against card fraud using a rules engine, the preventive screening and the screening of approved transactions are handled by fraud detection algorithms in the background. So, it's not purely rule-based; the AI algorithms are definitely involved as well. #00:07:52#

Interviewee All right, that makes sense now. I didn't read the part below. #00:08:14#

Interviewer And then for fraud alerting, the process is straightforward too. A fraud alert is generated, it gets investigated, and based on the feedback, if it's confirmed as fraud, further investigation is done. If it's not fraud, the case is closed. That's the process. But as you mentioned, the most interesting part is in the application layer, where everything comes together. We have the key components, starting with the fraud detection rules engine that handles the rule-based checks. However, the most important part is the fraud detection algorithm. This algorithm handles pattern recognition and anomaly detection for preventive screening and approved card transactions. It also plays a crucial role in reducing false positives, which is what you were referring to earlier. Another main component is the FDS frontend, where users interact with the system. This is where the interface is provided, allowing users to give feedback on alerts. Everything is visualized here, and the feedback collected from users is directly fed into the algorithm so it can learn and improve over time based on that feedback. Any questions? #00:08:18#

Interviewee Just one question. Maybe I missed what you said, but after performing the preventive screening, if the screening against the card rule hits, let's say the rule triggers for any amount greater than €10'000, does the flow proceed down from that point before the card transaction is blocked? #00:09:49#

Interviewer I mean, when there is a breach of the rule, the transaction is immediately blocked and doesn't proceed further to the approved card process. It stops there. So if the card screening identifies a breach, the transaction is blocked at that point. #00:10:23#

Interviewee Right before the block card transaction, there's a vertical line. I'm confused about that. What does the yes mean? No, that one, the one after it. Yes, that one. #00:10:46#

Interviewer So what this means is, if there's no breach of the rule, the flow moves to perform a preventive screening using the algorithms. If a certain threshold is exceeded, the card transaction is blocked. If the threshold is not exceeded, the transaction is approved. #00:11:00#

Interviewee I see, I was following the route above. So the flow will go up first, then come down? #00:11:22#

Interviewer Okay, yes, that's how it should work. The flow could definitely be improved to make it clearer, but essentially, an alert can be generated or the transaction can be blocked either by a breach of the rule or if the algorithm detects something suspicious. #00:11:28#

Interviewee I get your point now, that makes sense. #00:11:48#

Interviewer Exactly, it makes sense. Then we also have the support layer, which includes all the supporting elements like the data integration pipeline. This pipeline ensures that all the necessary data is collected from various sources, such as transaction data, contextual data, or historical data from our database. We also have a feature extraction engine that processes the raw data and extracts all the relevant features needed for further analysis. I mean, I won't go into too much detail about all the data flows, otherwise, we'll never finish. #00:11:50#

Interviewee It's very well explained, I understand it clearly. #00:12:34#

Interviewer We also have another important support component, like the monitoring system, where we track metrics and data quality. This includes checking what kind of data is coming in, whether there are any errors, missing values, or changes in the data. Additionally, we monitor for bias, for example, if a compliance officer provides inconsistent feedback, we need to see how the system reacts to that. Another key part is the stage model update system. This is where all updates, training, feature engineering, and threshold optimization take place. Importantly, this is done in a staging environment, not the live environment, so production is not impacted during testing or training. Once everything is finalized, it can then be deployed to the production environment. I hope that makes sense because you're actually the first person I'm explaining this to, so I hope it was understandable. #00:12:38#

Interviewee No, for sure. It definitely makes sense. To be honest, even if you hadn't explained anything, I would still have understood it. The whole flowchart and diagram are very well designed. I'm just checking the client database server to make sure everything is in place, and it looks very detailed. Nothing specific comes to my mind right now. #00:13:58#

Interviewer I mean, the only point is that the process flow here is a bit confusing with the arrows. It should more clearly indicate where the flow goes, and that probably needs to be adjusted. But considering the last time we discussed the design principles and everything, do you think it's all well integrated? #00:14:34#

Interviewee Definitely. I mean, for sure. You could add more details, like which messaging service is used and how it's integrated, though I'm not sure if that falls under your scope. But overall, this looks perfectly fine. If you want, you could also add an explanation engine, but that's up to you. #00:15:03#

Interviewer Apart from that, is there anything else? It always depends on the level you're focusing on, right? I'm not sure if I showed it earlier, but there's a dynamic compute cluster here. It's really basic, at a high level, to explain scalability and flexibility. You can add or remove nodes depending on transaction volume, like during peak times such as Christmas. If more power is needed, nodes can be added, or they can be removed when the volume decreases. Now, moving on to the second concept, it's more of a visual element. I created a simple dashboard for user interaction. I know you're more on the data science side, but I'm curious about your thoughts. This dashboard is designed to give users a quick overview of important metrics. For example, when you log in, you'd see information like card transactions over the last three months, system performance metrics like accuracy, flagged transactions, both approved transactions that were flagged afterward and blocked ones, and system alerts. These alerts could notify you, for instance, if accuracy falls below a set threshold, signaling that it might be time to check in with the developers or provider. It could also highlight an increase in false negatives, prompting the user to investigate further. The dashboard is customizable, users can add or remove widgets depending on their needs. For example, if someone isn't interested in accuracy, they can remove that widget and replace it with something like country distribution or any other relevant metric. It's meant to be a flexible and user-friendly tool. What's your opinion on that? Do you think it could be useful, or do you have any suggestions? #00:15:26#

Interviewee No, everything looks good. Just to clarify, the false negatives, false positives, and accuracy metrics can only be obtained after user feedback, right? Also, if a rule is triggered and the case is opened, but then the model auto-closes it, the agent won't check it, correct? I assume that's why it was auto-closed. So, wouldn't the false positive, false negative rates apply only to the anomaly detection model and not to the auto-closing model? #00:18:48#

Interviewer Yes, that could be an issue. Do you think the user should also review the auto-closed cases? #00:19:35#

Interviewee No, I don't think so, because the purpose of the auto-close model is to reduce the burden on users. #00:19:41#

Interviewer I mean, we can allow for some flexibility. For instance, if you look here, we have the blocked transactions, flagged transactions, or all card transactions. If I'm curious to see only the auto-closed cases, I can filter by the auto-close status. Then I can see, for example, this transaction was auto-closed because it was initially flagged for a high-risk country, but later auto-closed. I can view more details about the alert, including the status, auto-closed, and the AI confidence, which is key for transparency. We've discussed this a lot, the AI's confidence level in making the decision is shown, along with the explainable features. For instance, the case was auto-closed with a 91% certainty despite an identified risk. The explanation could be something like, the country matches the customer's usual behavior, perhaps because the customer works with a nonprofit that operates in high-risk areas. Now, if we switch to a blocked alert, say a recently blocked card transaction, you can see that the system blocked it due to a geolocation anomaly, again with a confidence score. What's crucial here is the feature impact, which shows which features led to the system's decision. For that, I used SHAP values, I think you or someone else suggested incorporating this, so you can really see that, for example, geolocation and timing were key factors. The context-specific map shows that the user typically transacts in Europe, but this time it was in Thailand, raising suspicion. It also includes risk score analysis, a summary of the alert, the client profile, and external data, providing a comprehensive overview for the user. What do you think about it? Is it too overwhelming, or does it seem manageable? #00:19:48#

Interviewee No, of course not. I understand it's not the final website, it looks a bit HTMLish, which is expected at this stage. But the level of detail is perfectly fine. #00:22:32#

Interviewer I mean, do you think something is missing? I tried to cover everything. #00:22:45#

Interviewee No, I don't think you missed anything. Again, I'm not really a UI guy. #00:22:49#

Interviewer I guess the most insightful part for you would be showing the feature impact using SHAP values. Or what would be the correct technical term? #00:22:56#

Interviewee The technical term for that is model explanations. Just a side note, I'm not sure if you're familiar with SHAP, but it's model-agnostic. Some explanation methods depend on the type of model you're using. For example, there are methods like DeepLIFT, which work only on neural networks, and more specifically on certain types, such as autoencoders or image processing neural networks, because they need to access the individual neuron values. On the other hand, SHAP is model agnostic, it only cares about the feature inputs and the model outputs, regardless of the type of model being used. It looks at the input and output and compares how changes in feature values affect the outcome. I thought I'd mention that just to give you a clearer picture for your presentation. #00:23:19#

Interviewer And what does Hawk use, or in your work, what do you typically use to show explicitly which feature had the most impact on a decision? #00:24:18#

Interviewee Which feature do we use? Sorry, I didn't quite understand your question. #00:24:30#

Interviewer What I mean is, when you're working at Hawk, and you get an alert that's flagged as suspicious, how do you typically show why it was flagged? What's the best approach for you to explain to the user the reason behind the suspicion? I know this might not be your main focus, but just curious about your perspective. And now, just clicking through the prototype to the dashboard, I know it's not visually stunning, but it should work. I've added some descriptions here, but those are more for the user's understanding, like what each section means. It's a small, interactive tool. Any final thoughts on this? #00:24:32#

Interviewee No, as a developer, it looks perfectly fine to me. The UI people might have some concerns, but from my perspective, it serves the complete purpose it's meant to. #00:25:34#

Interviewer Do you think anything is missing in terms of features or aspects? As I mentioned, one thing I haven't integrated yet is tracking the FDS performance over time using metrics. This would show how the fraud detection system's performance has changed, indicating when updates were made. For instance, if an update caused performance to worsen, that would be visible, and then if another update improved the model, that would also be tracked. This could be useful for developers to see how model performance evolves. What do you think? Does it make sense to integrate this, or do you think it's not really necessary? #00:25:49#

Interviewee I mean, there's no harm in adding additional features, right? Of course, if you put everything front and center, it can become overwhelming. But if the feature is available for those who want to use it, then that's perfectly fine. #00:26:34#

Interviewer Thank you so much, really. Any final thoughts on it? #00:27:06#

Interviewee All good. Best of luck with your thesis! #00:27:14#

User 1 at amnis

Date: October 17th, 2024

Time: 16:30

Place: Microsoft Teams

00:00:00 – 00:35:21

Interviewer Perfect. This third round, the second evaluation, or the third interview is about discussing the applicability of the design principles. If you remember last time, we talked about those principles, and I collected all the feedback from you and other experts. I used that feedback to refine the design principles. Based on those principles, the next step was to create a concept or prototype to demonstrate how applicable and useful these design principles are. They were initially quite abstract, so now we need to see how they can be integrated into a concept or prototype to test their usefulness in a practical environment. As you recall, we discussed several design principles, such as development, data integration, scalability, and flexibility. For the technical aspects, I created an enterprise architecture model, which is quite technical but was discussed with the experts. Today's topic, however, is more relevant to you as users. It's important to focus on how you can interact with such systems. We also talked about the need for transparent and user-friendly features, including context-sensitive and explainable features. This is what we're going to discuss today. First, I need to share my screen. I created an interactive prototype. While it may not be the most visually appealing, it's designed to showcase the functionalities and how these design principles can be applied in a real-world scenario. Imagine this is your web app or front desk for working with the fraud detection system. When you log in, the first screen you see is your dashboard, which offers different possibilities and options. For example, you might want to monitor your card transactions, how many were approved, how many were auto-closed in the last three months, and how many were flagged as suspicious or blocked. You might also want to track the performance of your fraud detection system, such as its accuracy, false positive rate, and false negative rate, and see how these metrics change over time. These features are configurable. For instance, if you want to add a widget to see the country distribution of your card transactions, you can easily do that. This customization caters to the needs of each company or user. At the bottom, you can focus on blocked transactions, which shows all transactions that were blocked before approval. You'll see how many alerts are new, how many are under review, and the flagged transactions that were analyzed again. You might also receive system alerts, such as if the accuracy falls below a certain threshold or if the transaction volume is significantly higher than usual. All these widgets serve as shortcuts, so if you want to check new alerts, you can quickly navigate to them. So, my question is, what do you think of this dashboard? How well does it integrate an overview of system performance, usability, and transparency? What are your thoughts on this concept? #00:00:07#

Interviewee Definitely, I like it. At first glance, it seems like you've built a strategy based on the insights you've gathered from our side, right? That's well improved, of course. It looks really nice. The basic details about the blocked transactions are essential. It's necessary to have them blocked to prevent any payments to sanctioned entities, like a Russian payment or any transaction associated with a geolocation anomaly. Those should remain blocked to ensure compliance. Overall, it looks pretty well enhanced and well-designed, so good job! #00:05:30#

Interviewer Thank you. If we go further, this should serve as the dashboard for you. It's obviously more interesting to see how you can handle a fraud alert, right? This is just to showcase how you could keep track of things and create more transparency. There are other parts that are not currently implemented or are not the main focus, but I want to highlight them shortly. Model governance, here, you can have an overview of how the system was trained, including ethical guidelines, data used for training, and a change log that details what has changed over time. You could also integrate a manual section where, as a manager or senior officer, you could directly create new rules or modify existing ones. If you add or edit a rule, this could enhance collaboration and support the design principle of continuous improvement. There could be a tab for direct communication with developers or data scientists, change requests, or a system to report issues, essentially functioning like a ticketing system. Additionally, basic export functionalities, allowing you to generate reports based on your criteria or export all relevant information. Before we dive deep into all the large-scale details, we could also have a performance overview that shows how the system's performance has changed over time. This allows you to track improvements or identify if something isn't going well. For instance, if an update labelled as version 2.0.3 was deployed and we see that accuracy went down, we could assess that this update wasn't effective, or there might have been an issue with threshold optimization. After the next update, if performance improves, it indicates that the update worked. This is all aimed at creating more transparency. So, as a user and senior officer, what do you think about such a tool? Would you find it applicable, or would you say you don't need that functionality? #00:06:43#

Interviewee Technically, it's okay. However, I might be thinking outside the box now, because the key factor here is whether fraudulent activity is identified or not, which ultimately proves the robustness of your setup and system. That's always critical because it depends on multiple factors. No platform can claim to be 100% accurate. The reality is usually quite different, it's more about assessing whether something is fraud or not and then investigating it further rather than simply declaring it as fraud. So, while discussing accuracy and scoring is nice, we have to question whether those numbers are real. For instance, what's considered normal, and what isn't? If a system is scoring a million transactions, we need to understand those metrics. If those metrics don't make sense, it undermines the system's reliability. What I've noticed is that your main focus here is on fraudulent activity, but there are elements being combined that don't necessarily relate to fraud. For example, if someone makes a payment to a sanctioned entity, that's not inherently fraud. It's more about transactional monitoring and compliance with sanctions rather than fraud itself. So, combining performance metrics with fraud detection can be a bit tricky. #00:09:18#

Interviewer You have to be careful about what you really want to track, so we need to ensure more transparency. As you mentioned, these blocked transactions might involve sanctioned countries or dual-use limits, and they may not necessarily be fraud cases. In the end, you can still look at whether a transaction was done, let's say, for example, with a card in Russia, which is currently sanctioned. It's possible that this card was stolen and then used in Russia, even though the client would never intentionally use it in a sanctioned country. It really depends on the context. #00:12:07#

Interviewee So then there's a scenario where the fraudster is from Russia. In that case, you can certainly apply it that way. The fraudster could attempt to steal money through a point of sale or an internet transaction with a counterparty based in Russia. #00:12:39#

Interviewer For example, we have a new transaction that was blocked, so let's click on it. I know there's a lot of information on the screen, but let's go through it. The status shows it's new and blocked by AI. The other, we see it was related to a sanctioned country and was blocked due to a location anomaly. Currently, it's unassigned, but you have the option to assign it, escalate it to a more senior officer, or resolve it. First of all, we see that the AI found a suspicion and is highly confident in its assessment. This adds to the transparency regarding the system's decision. Here, we can see that the AI confidence is quite high, stating that it's highly confident this card transaction is fraudulent. We can dive deeper to create more transparency and explainability. The feature impact shows that the location significantly influenced the decision, along with the timing and the behavior of the transaction. In a real-case scenario, there would be other features to consider depending on the specifics of the case. I'm just showcasing what impacts this decision. We also have the transaction information here and the merchant's name already sounds a bit suspicious, and it was done in Thailand. #00:13:09#

Interviewee It's definitely obvious that if I read the alert summary and see that the card was used in Switzerland because the cardholder is based there, then a transaction being processed in Thailand raises some red flags. There's an open question here, though. If I look at the transaction type, which typically isn't a point of sale, it suggests that it would be impossible to physically get from Switzerland to Thailand within 49 minutes to use the card. However, if it's an online transaction, technically, someone in Switzerland could access a website where the payment processing entity is based in Thailand. So, while it's clearly possible to make an online purchase from Zurich, it definitely warrants some attention and at least a closer look. If you're ordering something online, it's not impossible, but it's something that should be investigated further, especially if the transaction is flagged in this manner. #00:15:03#

Interviewer Looking at that map, it's a really good point. I never thought about that. You've already identified a flaw in my reasoning. #00:16:30#

Interviewee Because it's always about the card type, right? With point of sale transactions, it's pretty straightforward, the card is physically presented at the location. However, with online transactions, it could be processed from another part of the world while the cardholder is sitting in Zurich. So, it's really not that clear-cut in this case. #00:16:46#

Interviewer Right. Then let's say it was a point of sale transaction. #00:16:49#

Interviewee OK, so then it's clear, that's definitely fraud, right? #00:17:13#

Interviewer Then, I added a really context-sensitive feature here. You can swipe through all the information, like the user's last transaction volume and how it has changed over time. Currently, as we noted, it's about the geolocation problem. The system shows a map of the card transaction, indicating that the user typically shops in Europe. However, now there's a transaction in Thailand, which raises suspicion. This feature helps you by providing relevant information about what's going on. If you want more details, you can switch between views, but the current focus is on the geolocation issue. The system highlights this concern and guides you with the map. We also display the last transactions with the same card, along with the alert summary generated by the system, giving managers a quick overview without needing to dive deeper. The alert summary includes a risk score analysis, showing that the score is above the threshold, which is why the alert was triggered. Another good aspect of this concept is that it integrates

client data, so you don't have to switch between platforms. You can see the client profile directly here and dive into more details if needed. Additionally, it could pull external data from sources like sanctioned lists or public databases. Imagine incorporating a functionality similar to ChatGPT that can search for information on merchants like Bitconnect and flag any associations with cryptocurrency scams. This integration provides relevant information on one screen, eliminating the need to switch between platforms or search in Google or other databases. So, what do you think about the transparency, user-friendliness, and explainability of this approach? #00:17:15#

Interviewee I like it. That would be nice. Having that information readily prepared is exactly what you'd expect, especially given the context of a 49-minute transaction window. The key message here is understanding the relationship with entities like Bitconnect, if that's flagged as a cryptocurrency scam, it's crucial to highlight that in one go. I don't want to interrupt too much, but I had a really interesting discussion yesterday with the Hawk team. They pointed out the importance of cooperation between the fraud team and the AML monitoring team, which are usually separate. This collaboration is highly beneficial and adds significant value. However, there's something I want to mention. Many technical details of card transactions, which aren't visible here, involve major card providers like Visa, MasterCard, and American Express, which handle all the counterparties. If Bitconnect is clearly a troublemaker, the message is that the card networks would respond quickly, they would shut them down. But the challenge lies in the timing of such actions. You never know how long it might take to block a fraudulent account or remove a point of sale. It could be that they just started scamming last week, and this is the moment you discover the situation. It's probably too complicated to set up, but having all this external data at your fingertips would be essential. If I don't need to switch to Google to research who Bitconnect is or why it's flagged, that would be a great feature. That's cool. #00:19:50#

Interviewer If we go further, we can also take a quick look at tool alerts. Ultimately, you can resolve them, comment on them, and add evidence if it was handled as fraud. You can also assign it if it was not resolved or wasn't fraud. Additionally, you have the option to report an issue if you feel that an alert shouldn't have been created, stating that it makes no sense for that case. This demonstrates direct collaboration with the provider. Now, let's look at another alert. This one is different, it's new but blocked by a rule. So, it wasn't blocked by the system itself but rather by a specific rule. Here, we can see which rule was broken. In this case, it was due to a transaction from a sanctioned country, let's say Russia. The transaction occurred last year. Since it's a blocked transaction based on a sanctioned country, we don't need to see the AI confidence level here, it's clear why it was blocked. The alert shows a list of sanctioned countries for context, so you know why it was flagged. It also provides external data, indicating that Russia is on the FATF list and is designated a sanctioned country. This gives you more information about the situation. However, as you mentioned, it's important to differentiate between whether this is an alert for fraud or just a rule breach. Currently, it's treated as a fraud case, but you might argue that it doesn't necessarily indicate fraud, right? #00:23:28#

Interviewee It's sanctioned, but technically, you also need to be very careful when looking at your list of sanctioned countries. Just because a country is FATF-listed doesn't mean it's sanctioned. The terms are different, sanctions involve prohibitions, while FATF listings can include a wider range of countries. In fact, there are only a few countries that are outright sanctioned, like Iran and North Korea, for example. There might be others, like Croatia and South Africa, that could be considered problematic but don't necessarily fall under the same category as sanctioned countries. #00:25:28#

Interviewer I mean, the country list is just an example. #00:26:12#

Interviewee Exactly. Because if you travel to Croatia during the summer and get flagged for a transaction, like paying for ice cream, that's not the point of these sanctions at all. #00:26:15#

Interviewer It's all about context here. It doesn't make sense to show a map for this situation because it's not related to timing. Instead, it's more relevant to display a list of sanctioned or prohibited countries. If you have your own list of prohibited countries, that could be included as well. Now, if we look at all card transactions, you can filter for auto-closed transactions. For example, let's examine this one that was auto-closed by the system. As I mentioned, it was flagged by the AI, meaning it was initially an approved transaction that underwent a secondary investigation. In this case, the transaction was conducted in a high-risk country, let's say Haiti. This is why the alert was created, but the system closed it. We can see that the AI was highly confident that this case could be auto-closed based on the feedback you provided. We can also analyze why the AI is confident about this decision. #00:26:34#

Interviewee If they travel to Haiti and pay for a truck rental, that's definitely something we can accept. That's for sure. #00:27:52#

Interviewer Right. I mean, we can match the transaction with the customer's behavior. If it fits the pattern of their profile, we know that their last transactions were also in high-risk countries like Haiti, Mali, or Kenya. When we dig deeper, we see that the client type is a nonprofit organization focused on food supplies. They frequently operate in those countries to help people, so renting a truck would make sense in this context. From the client profile and historical data, we can determine that this transaction can be closed,

it has nothing to do with fraud. If it were a genuine fraud case, we could report it initially and flag it for review, noting that the system made a wrong decision in this instance. We'd remind it not to auto-close such cases in the future. In this case, the context-specific feature shows the high-risk country flagged for the transaction. This classification can change depending on the case, helping you understand the situation better. So, what do you think about the explainability and transparency aspects in real life? Is there anything you feel is missing? #00:28:02#

Interviewee What I'm really looking for when examining the last transaction is that quick overview. It's very important, of course. Even if it seems nonsensical, like one day in Basel, the next in France, and then in Kenya or Malaysia, that's just impossible. But when I look at the details, knowing where the transaction took place is crucial. It would also be helpful to include the merchant's name for additional context. However, having a quick overview of when the transaction occurred, the user's time zone, and where the card was used is exactly what you need to focus on. That's what's required, and it's similar to what we have in our Hawk system. We check those details early on by principle, so it's very nice. Technically, high-risk countries can be somewhat interchangeable. It would be good to highlight the specific country related to this alert, just for clarity. That would add value, but I see it's already mentioned in the alert summary. It says, watch for high risk, which is useful. Overall, I don't have any comments on how to make it better or any additional ideas. #00:29:26#

Interviewer Is there something missing that you feel you can't work without? Or do you think there's something negative about the current features? Is there a future capability or something that isn't well represented in what I showed you? #00:31:21#

Interviewee From a historical point of view, it's very important to look back and see if there has been any previous interaction with this counterparty. If they've made multiple payments already, that context would be helpful. For instance, if they travel there every month, it would be nice to see that historical perspective. Having that information would indicate that it's normal behavior. You could easily evaluate that, for example, the truck rental in Haiti has been used multiple times in the past 12 months, let's say 15 times or whatever the number is. That kind of data would add valuable context. #00:31:32#

Interviewer That would provide a more detailed view, allowing you to click on the client profile to see their recent activity. You could also have a map displaying all the transactions, so you can observe the flow and timing of those transactions. It would give you a comprehensive understanding of their behavior and interactions. #00:32:42#

Interviewee Exactly. It's again a very specific and extreme situation, like when an NGO is traveling heavily around the globe. That's not usually common practice, typically, the coverage of places is very limited. However, travel agencies and companies would definitely fit into that category. In those cases, it would be much easier for me because I'm focused on resolving fraud and alerts on a daily basis. If I see one transaction flagged in Haiti, I need to quickly determine if it's normal behavior. If they pay there every month, there's no problem. But if it's their first time, that raises a red flag, and I need to be careful. #00:32:59#

Interviewer Well, that makes sense. If you don't have anything else to add, then we are done. #00:34:18#

Interviewee I think, overall, it's important to focus on the specifics, particularly regarding the number of false positives and false negatives. I'm really considering what exactly the negative rate signifies. If something is flagged as negative, it means there's a problem, right? From the perspective of value explanation, that's crucial to understand. #00:34:29#

Interviewer Perfect, thank you. #00:35:13#

User 2 at amnis

Date: October 16th, 2024

Time: 13:30

Place: Microsoft Teams

00:00:00 – 00:36:23

Interviewer Now it works. Sorry about that. So, as I mentioned before, the next round is focused on another evaluation. #00:00:32#

Interviewee No worries. #00:00:45#

Interviewer How applicable those design principles are in real, practical environments is the focus here. Last time we discussed the design principles, and I've incorporated all the feedback I received from you. I've refined some design principles. Just as a reminder, I'll go through them briefly with you, but not in too much detail. Afterward, I'll show you how these design principles can be applied in a real-life scenario. On the technical and implementation side, I developed an enterprise architecture model, which I'll discuss with our experts from Hawk. However, for you, as compliance officers, it's more important to see how users interact with such fraud detection systems. So, I'll show you a showcase of my prototype for this system. Just as a reminder, we previously discussed the adaptive learning and continuous improvement design principle. One of the most important ones we'll discuss today is the user-centric development and alignment, along with comprehensive data integration and quality assurance, AI-driven and targeted rules, scalability, and flexibility. However, the one we're going to focus on today is the transparent and user-friendly explainability. This is essential for working with the system in a clear and user-friendly environment. Lastly, we have the balanced approach to fraud detection and bias mitigation, along with ethical decision-making. But as I mentioned, we'll focus on the transparent and user-friendly aspect today. For this, I'll show you a prototype built in Figma. It's an interactive tool, similar to a web-based tool you might use at Hawk. You log in, and the first thing you see is your dashboard, displaying important information. From the interviews and feedback I've gathered, incorporating widgets are a good choice. These widgets show relevant information and can be customized based on user needs. For example, in this showcase, I'm displaying card transactions over the last three months, along with a performance overview that shows the current system performance. Here, you can track metrics like accuracy, false positive rates, and false negative rates. There's also an info box that explains what these metrics mean if the user needs clarification. For compliance officers, one of the most important features is seeing the number of blocked transactions. You can see how many alerts are currently blocked, how many are under review, and also flagged transactions from post-hoc analysis. For example, there are 7 new alerts and 39 in review. These features are customizable. If you want to add a country distribution widget, you can do that, and it will display which countries have the most transaction volume, like Switzerland or Germany. You can also have a list of recently blocked card transactions or approved transactions for easy monitoring. The key here is functionality. So, what do you think about the functionality of this dashboard? It's less about how it looks and more about how useful it is for you. #00:00:59#

Interviewee I get that. I really like the fact that you can actually edit the dashboard yourself, making it adaptable to your specific needs. I can see how that could be very helpful. For example, if there are two different compliance officers with slightly different roles, one might focus on unblocking transactions, another might be more into data research, and a third could handle flagged transactions that are out of the ordinary. This feature would be really useful because you wouldn't need to assign different access rights to each officer. They would all have the same access, but with 12 or 20 different widgets or features to choose from. They could set up the dashboard based on their individual workflows. We've had a bit of a struggle with this recently, especially after our newest member, Magdalena, joined. She needs more detailed access to the data, but at the same time, she's more focused on the practical aspects, like closing alerts, rather than researching the data behind them. She's more of a front-end user, so this kind of customization would definitely help us. It allows each person to set things up the way they want, and that's really good. #00:05:31#

Interviewer Thank you. This is just a dashboard to show you some information, mostly focused on displaying metrics, which we've discussed several times. #00:07:22#

Interviewee Yes, it seems very user-friendly. #00:07:27#

Interviewer Now, as an example, we have a new alert for you to handle on the right. Just enter this one, and while I know the design isn't really pretty, let's focus on the functionality behind it. If you had to work with such an alert, let's start with the basics. The status is clearly marked as new. We know it's blocked by the AI, not by a rule, meaning it wasn't approved due to a suspicion the AI detected. In this case, the reason is a geolocation anomaly, meaning the transaction occurred in an unusual location. Currently, the alert isn't assigned to anyone, but we can also see the AI confidence level, which tells you how certain the system is that this is a fraud case. Here, the system is 91% confident that it's fraud. Another important aspect is explainability and transparency. You can see why the decision was made and which features influenced it. In this case, we can see that geolocation and timing, along with behavioral anomalies, had the most impact. This means the transaction didn't fit the usual pattern for this client. We also have the basic transaction details, but the key feature here is the explainability component, which shows you that this is a geolocation anomaly. The system provides a map, so as a user, you can assess whether the transaction seems fraudulent. For example, you can see that the client typically performs transactions in Europe, but this one was in Thailand, which raises suspicion. This is a simple case to illustrate the explainable feature. The map is provided because the issue is related to geolocation. You can see the last transaction occurred in Zurich, and now there's one in Thailand. There's also an alert summary for senior officers or managers who don't need to go into too much detail, offering a quick overview of the situation. For example, the summary might say that the card transaction in Thailand was executed 49 minutes after the last one in Zurich, and it could include external data, like the fact that the merchant has been flagged in reports related to cryptocurrency scams. The system integrates external data as well. You can also see a risk score analysis, showing you that the transaction exceeded a threshold and was flagged. Given all that, what do you think? As a user, when it comes to user-friendliness, explainability, and transparency, do you feel the design principles are well incorporated here? Is there anything missing, or do you think something could be improved? #00:07:31#

Interviewee I mean, I think all the necessary information is implemented in this overview. The basic analysis can be done from here, and you can dive deeper if needed. From a practical standpoint, the only thing that stood out to me at first glance was the options in the top right corner to assign, escalate, and resolve. I'm wondering, until you actually assign the case to yourself, should you even be able to submit a resolution? That's just a thought. From my own daily work, typically, you can't resolve, reject, escalate, or approve anything until you've assigned the case to yourself. I understand that in a comprehensive overview like this, when you can't go deeper, it might make sense that after assigning the case, only then should you be allowed to resolve it. That was just one of the first things I noticed. But this is just my personal opinion, and I guess everyone has different preferences. I like how the data is all in one place and compact, but I could see myself struggling a bit to navigate the page. It might feel a bit overwhelming with so much data displayed at once. #00:11:40#

Interviewer True, it's a lot of data in one place. #00:13:37#

Interviewee Yes, maybe structuring it differently could help. For example, a dropdown option could be useful for some elements, like the map in the middle, it feels a bit overwhelming. The map is a great idea because a visual representation can make things click instantly. For instance, if you can see that transactions happened in Geneva, Lyon, and then suddenly one in Thailand, it immediately raises a flag. So, the visual aspect is important. However, I would suggest making the map an optional element, like a dropdown where you can choose to view it if needed. Maybe you could have the recent transactions displayed first, and then an option to expand and see the map. That way, it's not as overwhelming right away. In terms of how I would approach this, I first need to identify who the customer is, why the transaction was flagged, and how it differs from the rest. I think all that information is already there, I can see why the alert was generated, the AI's confidence level, the basic transaction details, and a summary of the customer. So, overall, it has everything. #00:13:42#

Interviewer It's overwhelming. Maybe having a dropdown option or a switch to toggle between different views, instead of displaying everything at once, could help. #00:15:30#

Interviewee Something like that. Exactly. #00:15:35#

Interviewer That's a really good point. Now, moving forward, we can resolve the case by adding comments or evidence if we have it. We can also specify if it's fraud and what kind of fraud it is to help the system improve with correctly labelled data. Additionally, if the alert seems completely out of context, you can report an issue. Otherwise, once it's blocked, you can assign it to yourself, escalate it, or resolve it, those actions should happen before final resolution. If we go back, we can look at another alert. This one is blocked by a rule. You can see that instead of AI confidence, it's flagged due to a rule violation, specifically, because it involved a sanctioned country. Instead of a map in this case, it's about explaining context-sensitive features, so it shows the list of sanctioned countries and why this rule triggered the alert. I think you're right though, instead of displaying everything on one page, maybe this central section could be more interactive with tabs or a dropdown, so it doesn't feel as overwhelming. #00:15:40#

Interviewee I see your point. It's a great feature to have, especially for someone in compliance who is familiar with which countries are on different sanction lists. The updates for these lists aren't that frequent, maybe

every three months for the main ones like European and American lists. But sometimes you come across a transaction in a country where you don't typically handle transactions, and you have to check external sources like OFAC or FATF for relisted or blacklisted countries. Having that list within the system would definitely be helpful, but similar to the map, it could be an optional feature. Maybe have it at the bottom or as a toggle. For example, if the transaction is with Russia, the system can flag it as a sanctioned country, and you wouldn't need to manually confirm that it's on the list. Another thought—if the system already flags the country as sanctioned or blacklisted at the transaction level, I don't think you'd need to see the full list of sanctioned countries for every transaction. That could be something more useful in the settings, where users can adapt the risk level of each country based on the company's internal policies. Not every company follows official lists exactly; they might have different approaches to specific countries. So, having the option to customize that in the settings would be more helpful than seeing it with each alert. When it comes to rule-based alerts, the most important information should be what kind of transaction it was and which rule was breached. If a specific rule is breached, the system gives you the relevant details, but you don't need to go into as much depth as you would with an AI-based alert. #00:17:15#

Interviewer Exactly. It really depends on the case. Each one needs to be handled differently, whether it's an AI-based alert, rule-based, or an auto-close alert. #00:20:18#

Interviewee I understand. For a transaction in a sanctioned country, the most important details would be the transaction specifics. Even the map might be useful in this case, showing that recent transactions were done elsewhere, but suddenly there's a transaction in a high-risk jurisdiction. It could be flagged as a potential fraud attempt and then checked with the customer. From what we saw on both of the alerts, I feel like no important information was missing, but maybe some of the information displayed on the main screen wasn't necessary. It's more about having the option to view certain details if needed, but not having them all on level one when you first open each alert. That's more of a personal preference, but I think having a cleaner, less cluttered view at first would be helpful. #00:20:24#

Interviewer It's important to work on different levels, especially when it comes to the detail level. #00:21:42#

Interviewee Exactly. Sometimes you can resolve an alert just from level one, like seeing a payment and deciding it's fine, releasing it. Other times, you can't make a decision based on the overview alone, so you need to dig deeper, checking sanctioned countries, looking at the map, and so on. It's about working from top to bottom as needed. That's the approach I'd go for. But overall, I believe I could make a decision from the data provided in those two alerts. Good job, it has everything it should. #00:21:49#

Interviewer I mean, going further, we also have a list, for the blocked alerts, where we can filter for new, in review, or resolved alerts. We also have export functionality, which allows you to export or filter the data as needed. For flagged transactions, you have a list, and similarly, for all card transactions, you can filter as well. For example, here I've filtered for the auto-closed ones. Take this one as an example. The status is auto-closed. Initially, it was triggered by a high-risk country, so the transaction was flagged due to being in a high-risk location. However, if we go deeper, we see the AI had high confidence, 91% sure, to auto-close this alert based on the feedback provided or the behavior of the card usage. In this case, the analysis shows that the country matches the customer's usual behavior, likely because it's an NGO that operates in high-risk countries, providing aid or support. #00:22:33#

Interviewee Based on previous results, right? #00:23:39#

Interviewer Right. Or based on the customer profile, because we know it's an NGO, and they've already provided us with the information that they frequently do business in these high-risk countries. The transaction matches the recent pattern of their activities, so it's recognized as normal behavior. What do you think about the explainability feature here? Does it help you understand why it was auto-closed? I know you have access to the basic data, but I guess you'd prefer having different levels of detail. You might want to start with the fact that it was auto-closed, but then have the option to dive deeper into the specifics. It should follow a layered approach, right? #00:23:41#

Interviewee Exactly. The list of countries and the analysis are crucial, especially for closed cases. You need to understand the reasoning behind the AI's decision. If you can't determine why the decision was made at first glance, it's hard to know if it was correct or not. That's when a compliance officer would need to step in and provide different input to adjust the learning curve for the AI. I'm glad you find it useful that we've implemented this here. I had to laugh a bit too because, for example, Hawk doesn't have this feature. For card transactions in high-risk countries, every transaction triggers an alert because it's rule-based, and the AI isn't yet applied for automated closure. So, even if we know it's a travel agency giving their guides cards to pay for safari events or other services in high-risk areas like Central Africa, every transaction gets flagged as suspicious, not blocked, but still flagged. We then have to manually close every single alert, and you can imagine how quickly that adds up. I also think it would be great to implement something like a visual threshold indicator in the data analysis. For example, if a geographical rule scored 8 out of 10 points, it would be useful to see what the threshold is for each category. If I see 8 out of 10 but know that 7.9 is the threshold, I'd want to know that we're on the line and might need to pay closer attention. Having a simple horizontal graph with different colors to show how far away the

score is from the threshold would be really helpful in quickly assessing whether an alert is borderline or if it's well within the normal range. #00:24:22#

Interviewer That's a really interesting and good point. I need to mention that while it's not currently integrated, there would be functionality in the FDS to export data, create reports for audits, or export transactional data to analyze system performance. What I think could be particularly useful is having a performance overview over time. You'd be able to see how changes, like threshold optimizations or the implementation of new rules, impacted the system. For instance, if an update led to worse performance, you'd be able to identify that and see when a new update improved the system again. This is more on the technical side, and it might be something a manager would be more interested in than a compliance officer, but still, it's useful to have that data available. It could also be a good indicator for compliance officers, if false negatives start to rise, it might be a signal for you to raise an issue and flag that something is wrong. #00:26:56#

Interviewee Of course, I don't see all the filters. #00:28:44#

Interviewer Yes, it is just an example. #00:28:51#

Interviewee Probably, yes, because for me, the ability to change which rule you're focusing on is really important, especially when we're talking about rule-based instances. Being able to switch between specific rules would be key, since looking at the overall result of all rules combined doesn't provide much insight. That would likely fall under the filters, not just time frame, but maybe within the metric part as well. That's just the first thing that came to mind. I believe this part could be something to go over with someone from compliance, who, as you know, really dives into the monitoring of alerts and data improvements. He's more focused on how to adapt and improve over time. I don't have much better input for you regarding this, but I do think it's important to track improvements, or declines, over time. However, I'm probably not the best person to provide examples or suggest additional helpful tools for this. #00:28:52#

Interviewer Another part that isn't incorporated yet but is possible is the rule management section. This would allow users to add or modify rules, such as adding countries or setting limitations, all within the fraud detection system. Additionally, there's the model governance part, where users could see all the documentation and transparency related to the model, such as when it was trained, which updates were made, and what kinds of changes were introduced. There's also the potential to show ethical guidelines, which was brought up in some discussions, to explain how the data was selected and which features were used or excluded. You'd also have a change request section, allowing users to report issues with alerts, request changes, and interact directly with developers or the provider. This would include status updates on tickets, showing what's been done and any open bugs. These features aren't implemented yet, but it's important to show that they can be integrated into the system. So, overall, what do you think? #00:30:09#

Interviewee I see. I'm satisfied with how everything looks. I'd say it's very well done. Just one last thing that came to mind. I noticed the notifications settings in the top right corner. What I'm about to mention may not be directly applicable to your example, but I thought I'd share it since it came up in a webinar I attended on similar topics. There was a feature presented that could be useful, particularly for larger corporations with more employees. It was the option to follow a customer. Essentially, even if a transaction didn't get flagged as suspicious, the compliance officer or the person assigned to the case would still receive notifications for all activities related to that customer. The added value comes from being able to keep an eye on specific customers, especially in situations where portfolios are split among many compliance officers, like in fintech companies. If, for example, an officer is managing a smaller group of customers, they could receive real-time notifications on the transactional activity of those customers. So, while working on a case, a notification would pop up, letting the officer know that Company XYZ made a transaction, allowing them to check it immediately. It's on a different scale, but I thought it was interesting and relevant since we're discussing similar systems. I saw it last week and figured it was worth mentioning. #00:31:25#

Interviewer And obviously, it's a great feature when you want to keep an eye on a specific client, to follow their transactions and monitor their activity. #00:33:32#

Interviewee Exactly. You can track the customer closely. For instance, at a higher level, you can always do research using filters to isolate transactions based on the customer's name or ID, allowing you to see all their history. This feature would provide a very granular, detailed approach for specific customers who have triggered fraudulent activity in the past or are more exposed to potential fraudulent attempts. #00:33:45#

Interviewer Right. You may also want to follow specific card users, not just the account as a whole, but individual card users by name. #00:34:20#

Interviewee Exactly. Maybe you could implement tracking for a specific card, not just a specific customer. You probably have ideas for that. #00:34:28#

Interviewer So, in the end, what do you think? Are these design principles well implemented overall? Does it work as intended, or do you think something is missing? Does this principle completely not match the expectations for such a system? #00:34:41#

Interviewee It definitely does. I'm positive it would work in a real environment on a basic level, based on what I've seen. Apart from the visual aspect, there are some very interesting features. The widget functionality, where you can select anything you want, really struck me because navigating the platforms we currently use can be quite tricky. Even with the new compliance officer on our team, she has a lot of questions about where to find things, and sometimes I struggle to navigate as well. So, being able to customize the dashboard like this is really cool and something I genuinely appreciate. I would say that the system would succeed in a real environment. #00:35:01#

Interviewer Thank you so much. #00:36:05#

Interviewee No worries. #00:36:11#

User 3 at amnis

Date: October 24th, 2024

Time: 16:30

Place: Microsoft Teams

00:00:00 – 00:20:20

Interviewer In dieser dritten Interviewrunde, oder besser gesagt der zweiten Evaluationsrunde, möchte ich an unser letztes Gespräch anknüpfen. Damals haben wir uns die Designprinzipien angesehen, und ich habe Feedback von dir dazu eingeholt. Diese Prinzipien basierten auf einem Framework mit acht Punkten. Kurz zusammengefasst ging es dabei unter anderem um adaptives Lernen, Datenintegration, Flexibilität und Transparenz sowie Benutzerfreundlichkeit. Ich habe das Feedback von dir und den anderen gesammelt und daraufhin die Designprinzipien verbessert und erweitert. Der nächste Schritt in meiner Arbeit bestand dann darin, diese Designprinzipien in der Praxis umzusetzen. Hierbei habe ich zwei unterschiedliche Konzepte erarbeitet. Zum einen ein Architekturmodell, das sich stärker auf die technische Implementierung konzentriert, und über das ich vor allem mit Hawk gesprochen habe. Zum anderen ging es darum, mit den Usern zu erörtern, wie ein Prototyp aussehen könnte, oder wie ein solcher in einem realen Szenario implementiert werden kann. Dafür habe ich einen Prototyp erstellt, den ich gerne mit dir besprechen würde. Dabei liegt der Fokus vor allem auf Transparenz und Benutzerfreundlichkeit. Falls du dazu Fragen hast, können wir gerne darüber sprechen. #00:00:13#

Interviewee Ja, passt. #00:0:34#

Interviewer Genau, und hier möchte ich dir jetzt den Prototyp zeigen. Das ist der Prototyp für das Dashboard, welches ich erstellt haben. Sobald sich der User einloggt, sieht er direkt das Dashboard mit den wichtigsten Informationen, die alle konfigurierbar sind. Du als Manager kannst also entscheiden, welche Informationen für dich relevant sind. Beispielsweise könntest du weniger daran interessiert sein, wie viele Alarme gerade eingehen, sondern mehr daran, wie viele Kartentransaktionen in den letzten drei Monaten überprüft, blockiert oder als verdächtig eingestuft wurden. Das Dashboard bietet verschiedene Widgets, die vollständig interaktiv sind. Zum Beispiel könntest du schnell überprüfen, wie die aktuelle Systemleistung ist. Ein weiteres Beispiel wäre die Konfiguration eines Widgets, das die False Positive Rate zeigt. Diese Widgets sind also nicht nur hilfreich, sondern auch komplett anpassbar. Du kannst selbst entscheiden, welche Informationen für dich relevant sind, wie z. B. deine Top 3 oder Top 4 Länder. Du kannst jederzeit neue Widgets hinzufügen oder bestehende entfernen, je nachdem, was du als Nutzer benötigst. Für den Compliance Officer sind natürlich andere Informationen wichtiger, wie z. B. blockierte Transaktionen, die nicht überprüft wurden, oder verdächtige Kartentransaktionen, die zunächst genehmigt wurden, aber in der nachträglichen Analyse noch als problematisch erkannt wurden. Auch eine Übersicht über kürzlich blockierte oder geflaggte Transaktionen ist hier integriert. Zusätzlich gibt es die Möglichkeit, verschiedene Berichte zu exportieren oder automatisiert Reports zu erstellen. Eine detaillierte Übersicht der Systemperformance ist ebenfalls vorhanden. Das System bietet auch die Möglichkeit, Regeln zu managen. Zum Beispiel können Regeln direkt von den Compliance Officers ohne technische Unterstützung angepasst werden, z. B. bei neuen Sanktionsländern. Die Model Governance ist ein weiterer wichtiger Bereich, in dem dokumentiert wird, wie die Modelle trainiert wurden, welche Daten verwendet wurden usw. Es gibt auch ein integriertes Ticketsystem, mit dem Nutzer Bugs melden, Change Requests einreichen oder anderweitig mit den Entwicklern kommunizieren können. Was hältst du von diesem Dashboard? Denkst du, dass es durch seine Transparenz und Benutzerfreundlichkeit den Usern helfen wird, effektiv mit dem System zu arbeiten? Oder gibt es Bereiche, die weniger interessant erscheinen? #00:02:36#

Interviewee Das sieht wirklich vielversprechend aus! Die wichtigsten Parameter sind definitiv abgedeckt. Was noch spannend wäre, ist ein zusätzlicher Qualitätsparameter, der die Effektivität des Systems auf einer höheren Ebene widerspiegelt. Es wäre hilfreich, einen Überblick darüber zu haben, wie viele Transaktionen effektiv erkannt wurden, im Vergleich zu den Fällen, die das System möglicherweise übersehen hat und später durch den Kunden oder das Business als verdächtig gemeldet wurden. Diese Metrik würde eine Art Gesamtqualitätsindikator darstellen und könnte dazu beitragen, die Glaubwürdigkeit des Systems von einer übergeordneten Perspektive aus zu belegen. #00:06:10#

Interviewer Wir könnten das definitiv so umsetzen. Zum Beispiel, wenn wir uns die Performance-Sektion genauer ansehen, sehen wir aktuell zwar nicht genau die Parameter, die du erwähnt hast, aber ich habe hier einen Filter gesetzt, mit dem ich die Accuracy, die False-Positive-Rate und die False-Negative-Rate über einen bestimmten Zeitraum anzeigen kann. Es wäre auch gut möglich, genau die von dir angesprochene Metrik hinzuzufügen, indem wir eine zusätzliche Grafik integrieren, die das Verhältnis zwischen den vom System erkannten und nicht erkannten Transaktionen im Zeitverlauf darstellt. Aktuell habe ich nur diese

drei Metriken ausgewählt, da sie für das Unternehmen besonders relevant sind, aber das Transaktionsvolumen oder weitere relevante Indikatoren liessen sich problemlos ergänzen. So könnten wir es wirklich an die spezifischen Bedürfnisse anpassen und sicherstellen, dass die wichtigsten Performance-Indikatoren jederzeit sichtbar sind. #00:07:30#

Interviewee Genau, das wäre eine wichtige Validierung der Performance. Wenn wir die Modellberechnungen direkt mit den tatsächlichen Daten aus der realen Welt vergleichen, ergibt sich eine Art Belastungstest für das Modell. Dadurch könnte die Glaubwürdigkeit der modellbasierten Parameter deutlich gesteigert werden, vorausgesetzt, die Ergebnisse der Modellparameter werden durch die realen Gegebenheiten bestätigt. Dieses Vorgehen würde es uns ermöglichen, die Zuverlässigkeit der Modelle und ihrer Vorhersagen besser zu beurteilen und gegebenenfalls Optimierungen vorzunehmen, falls Abweichungen auftreten. #00:08:24#

Interviewer Genau, und ein weiterer Grund, warum ich diese Metriken ausgewählt habe, ist die Transparenz, die dadurch entsteht. Zum Beispiel sehen wir hier, dass ein Update zur Version 2.3 durchgeführt wurde, und können nachvollziehen, wie sich dies auf die Accuracy sowie auf die False-Positive- und False-Negative-Rates ausgewirkt hat. So lässt sich im Detail beobachten, welchen Einfluss verschiedene Updates auf die Systemleistung hatten. Das gibt uns wertvolle Einblicke und ermöglicht es, gezielte Anpassungen vorzunehmen, wenn ein Update die Performance in bestimmten Bereichen negativ beeinflusst hat. #00:08:57#

Interviewee Sind die Updates hier Modell-Updates, Regel-Updates oder beides? #00:09:31#

Interviewer Guter Punkt. Grundsätzlich konzentrieren sich die Updates in diesem Bereich auf das Modell selbst. Wenn du oben auf das Fragezeichen klickst, gelangst du zum Model Governance-Bereich, wo die verschiedenen Updates detailliert beschrieben sind. Dort kann man genau sehen, welche Anpassungen am Modell vorgenommen wurden. Regel-Updates sind hier jedoch nicht enthalten, da sich diese Sektion ausschliesslich auf Modellanpassungen beschränkt. Vielleicht wäre es sinnvoll, in Zukunft auch Regel-Updates transparenter zu machen und separat anzuzeigen. #00:09:36#

Interviewee Interessant ist tatsächlich, dass nach dem zweiten Release in Version 2.3.01 die Kurve wieder ansteigt. Das zeigt, warum es so wertvoll ist, die Auswirkungen jedes Updates auf die Systemperformance im Blick zu behalten. #00:09:58#

Interviewer Ja, genau. Man könnte hier tatsächlich auch verfolgen und anzeigen, wenn eine neue Regel hinzugefügt wurde oder andere Anpassungen vorgenommen wurden. Das wäre besonders nützlich, um zu sehen, wie sich diese Änderungen auf die Performance und Ergebnisse auswirken. Es zeigt auch, dass Nutzer vielfältige Anpassungen entsprechend ihrer Bedürfnisse vornehmen können, je nachdem, ob sie als User, Entwickler oder Compliance Officer spezifische Anforderungen haben. #00:10:11#

Interviewee Ja, absolut, das ist ein wichtiger Punkt aus Governance-Sicht. Für die Validität und Glaubwürdigkeit des Systems ist es entscheidend, die Modellkennzahlen kontinuierlich mit realen Daten zu vergleichen. Wenn man sieht, dass die Modellprognosen auch in der Praxis gut korrelieren, stärkt das das Vertrauen auf Management-Ebene erheblich. Für tiefere Analysen, wenn es etwa zu Unregelmässigkeiten in den Kennzahlen kommt, ist es hilfreich, transparenten Zugang zu den spezifischen Änderungen zu haben. Wenn klar ersichtlich ist, dass eine Anpassung vorgenommen wurde, die beispielsweise die Accuracy beeinflusst hat, lässt sich der Zusammenhang viel leichter nachvollziehen. Diese Transparenz ist nicht nur für das Verständnis der Änderungen wichtig, sondern auch, um Vertrauen und Nachvollziehbarkeit zu gewährleisten. #100:0:34#

Interviewer Wenn wir jetzt mal zurückgehen und das Dashboard aus der Perspektive eines Managers betrachten, der sich schnell einen Überblick verschaffen möchte, können wir einen neuen Fall ansehen. Ich weiss, es sind auf den ersten Blick viele Informationen, und wir haben auch schon Feedback zu Dropdown-Menüs erhalten, um die Übersichtlichkeit zu verbessern. Als Manager interessiert dich wahrscheinlich vorrangig die relevanten Informationen, ohne zu tief in die Details eintauchen zu müssen. Das System bietet daher nicht nur reinen Text, sondern auch eine prägnante Zusammenfassung. In diesem Beispiel sehen wir, dass eine Transaktion von der KI und nicht durch regelbasierte Vorgaben blockiert wurde. Die KI hat die Transaktion aufgrund der Geolokation als potenziell problematisch eingestuft. Konkret wurde die Transaktion in Thailand ausgeführt, nur 49 Minuten nachdem eine andere Transaktion in der Schweiz stattfand. Das System hebt also automatisch ungewöhnliche Muster hervor, die ein Indiz für mögliche Unregelmässigkeiten sein könnten. #00:12:06#

Interviewee Ich verstehe es bereits beim Hinsehen, das ist eine sehr gelungene Darstellung. Genau so etwas möchte man haben. #00:13:24#

Interviewer Klar, es mag auf den ersten Blick viel Information sein, aber das Ziel ist ja, auf einen Blick die wichtigsten Daten bereitzustellen. Da wir hier ein kontextspezifisches Feature haben und es um eine geografische Anomalie geht, wäre es wenig sinnvoll, allgemeine Informationen wie das Transaktionsvolumen des Kunden über einen Zeitraum anzuzeigen. Stattdessen stellt das System automatisch die relevanten Details

bereit: eine Karte mit den letzten Transaktionen und der neuen verdächtigen Transaktion, die sofort ins Auge fällt. #00:13:36#

Interviewee Ich verstehe, worum es geht und was das Problem ist. Mit diesen Metadaten, wie etwa Bitconnect und anderen Informationen, wird klar, warum das System Alarm schlägt. Ich denke, das ist selbsterklärend. Wenn ich auf diesen Link klicke, habe ich sofort ein grundlegendes Verständnis davon, worüber wir sprechen. Es ist selbsterklärend und wirklich perfekt umgesetzt. #00:14:06#

Interviewer Ein Aspekt, der viel diskutiert wurde, ist das ständige Wechseln zwischen Plattformen. Du hast es gerade erwähnt, eine Integration unserer internen Datenbank könnte das erleichtern, sodass Kunden- und Transaktionsdaten direkt im System eingesehen werden können. Auch die Einbindung externer Datenquellen, wie etwa Daten von Drittanbietern oder spezifische Datenbanken über verdächtige Händler, wäre hilfreich. Dies würde die Arbeit vereinfachen, da die Nutzer nicht mehr zwischen verschiedenen Systemen wie Google oder anderen Plattformen wechseln müssten. Ein weiteres Feature ist das AI Confidence- und Feature Impact-Tool, das zusätzliche Transparenz schafft. Je nach Alarmtyp passt das System die Ansicht an. Zum Beispiel zeigt es bei einem Regel-basierten Block eine andere Übersicht als bei einem KI-basierten Alarm. Hier sieht man klar den Grund, z.B. dass die Transaktion in einem Sanktionsland stattfand. Das System passt sich also kontextabhängig an und zeigt je nach Art des Alarms die relevanten Informationen an. Dadurch können Manager ohne tiefere Detailanalyse schnell verstehen, worum es geht. Ein weiteres Beispiel ist ein Know-Your-Customer-Fall, bei dem die Ansicht erneut angepasst wird und erklärt wird, warum das System einen Alarm geschlossen hat, ohne dass ein manuelles Ticket erstellt wurde. Hier könnte das System etwa zeigen, dass das Land des Kunden nicht mit seinem Profil übereinstimmt oder dass bestimmte Transaktionsmerkmale von der letzten Aktivität abweichen. Hast du dazu Anmerkungen oder Vorschläge? #00:14:37#

Interviewee Ja, wie gesagt, ich sehe hier eigentlich nichts, was fehlt. Es wirkt auch ein wenig wie ein Experten-Dashboard, oder? #00:14:37#

Interviewer Fehlt dir aus der Sicht eines Managers irgendetwas Kritisches? Du arbeitest ja nicht ständig direkt mit den Alarms, aber gibt es aus deiner Perspektive vielleicht ein Ziel oder eine Funktion, die noch integriert werden müsste, um das Dashboard wirklich vollständig zu machen? #00:17:04#

Interviewee Bei der False-Positive-Reduktion sehe ich, das ist eher ein sekundärer Aspekt. Bei Sanktionsprüfungen ist es entscheidender, die relevanten Datenpunkte schnell zur Verfügung zu haben, und das scheint hier gut abgedeckt zu sein. Die False-Positive-Rate ist für das Management schwer zu bewerten, da diese Validierung eher auf der operativen Ebene stattfinden sollte. Die Konfiguration des Modells spielt hier eine zentrale Rolle. Es hängt stark davon ab, welche Schwellenwerte und Toleranzen im Modell festgelegt sind. Das Modell-Setup beeinflusst die Qualität der Ergebnisse erheblich, und aus meiner Sicht ist die sinnvollste Validierungsmethode der Abgleich mit realen Vorfällen. Wenn es zu einem nicht erkannten Betrugsfall kommt, sollte die False-Positive-Rate natürlich kritisch hinterfragt werden. Für einen einzelnen Fall ist das allerdings schwer interpretierbar und könnte eher auf ein systematisches Problem hinweisen, das gezielt analysiert werden muss. #00:17:19#

Interviewer Genau, das passt auch gut zu dem Feedback, das ich von Hawk und auch von Compliance erhalten haben. #00:19:07##

Interviewee Ja, genau. Das Dashboard fördert die Transparenz, indem es zeigt, warum das Modell so entschieden hat, wie es entschieden hat. Falls sich eine Transaktion im Nachhinein tatsächlich als Betrug herausstellt, ist es wichtig, nachzuvollziehen, warum das Modell nicht darauf angesprungen ist. In solchen Fällen ist es entscheidend, dass man den Entscheidungsprozess des Modells versteht und entsprechende Anpassungen vornehmen kann. Gerade diese Einsicht in die Modellentscheidung ist also wichtig, um gezielt Korrekturen einleiten zu können. #00:19:12#

Interviewer Gut, dann sind wir soweit durch. Falls sonst keine weiteren Punkte mehr offen sind, hätten wir damit alles abgedeckt. #00:19:48#

Interviewee Ja, ich möchte nochmal betonen, dass ich das wirklich klasse finde, tolle Sache, tolle Arbeit. Ich bin gespannt darauf, dass du mir die Technologie mal näher zeigst; das könnte auch in einem anderen Kontext interessant sein, um sie dort einzusetzen. #00:19:55#

Interviewer Vielen Dank, danke dir! #00:20:18#